



**Generation Interconnection
System Impact Study Report
for
Queue Project AE2-288
SUNBURY 500 KV
206 MW Capacity / 90 MW Energy**

February 2020

1 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

2 General

The Interconnection Customer (IC) has proposed an uprate to an existing Natural Gas generating facility located in, Shamokin Dam, Snyder County, Pennsylvania. AE2-288 project requests an increase to the install capability of 90 MW (206 MW Capacity) to the previous AA2-182/AB2-074 queue requests. The installed facilities will have a total capability of 1233 MW with 1233 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is May 1, 2023. **This study does not imply a TO commitment to this in-service date.**

The table below provides a summary of the project capability:

	MFO (MW)	CIR (MW)	MWE (MW)
Existing AA2-182	1030	977	1030
Proposed AB2-074 Increase	113	50	113
Proposed AE2-288 Increase	90	206	90
Total	1233	1233	1233

Queue Number	AE2-288
Project Name	SUNBURY 500 KV
Interconnection Customer	Shamokin Dam LLC
State	Pennsylvania
County	Snyder
Transmission Owner	PPL
MFO	1233
MWE	90
MWC	206
Fuel	Natural Gas
Basecase Study Year	2022

2.1 Point of Interconnection

AE2-288 will interconnect with the PPL transmission system as an uprate to AA2-182/AB2-074 via a new breaker position at the Sunbury 500kV substation.

2.2 Cost Summary

The AE2-288 project will be responsible for the following costs for the physical interconnection:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrade	\$ 0
Non Direct Connection Network Upgrades	\$ 0
Total Costs	\$ 0

In addition, the AE2-288 project may be responsible for a contribution to the following costs for Network Upgrades to mitigate overloads identified in this report:

Description	Total Cost
System Upgrades	\$ 42,971,701

3 Transmission Owner Scope of Work

There is no additional Transmission Owner scope of work for the physical interconnection of AE2-288. The Transmission Owner scope of work for the physical interconnection is captured in the AA2-182 System Impact Study report.

3.1 Attachment Facilities

None.

3.2 Direct Connection Cost Estimate

None

3.3 Non-Direct Connection Cost Estimate

None

4 Interconnection Customer Requirements

4.1 PPL EU Interconnection Requirements

PPL EU applicable technical standards that address requirements for interconnection of generation, transmission, and end user facilities can be found at the following link:

<https://pjm.com/planning/design-engineering/to-tech-standards/private-ppl.aspx>

4.2 IC Direct Transfer Trip (DTT) Requirements

PPL EU will require an independent communication path, for Direct Transfer Trip (DTT) of the IC Intertie Protective Relaying (IPR) Fault Interrupting Devices (FIDs). There are no additional requirements above and beyond the requirements already identified for AA2-182.

5 Revenue Metering and SCADA Requirements

5.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

5.2 PPL Requirements

Installation of revenue grade Bi-directional Metering Equipment will be required in the vicinity of the POI to measure kWh and kVARh. PPL EU will design and supply the required metering equipment; all installation costs would be borne by the IC including CTs/PTs. All metering equipment must meet applicable PPL EU tariff requirements as well as being compliant with all applicable requirements of the PJM agreements. The equipment must provide bidirectional revenue metering (kWh and kVARh) and real-time data (kW, kVAR, circuit breaker status, and generator bus voltages) for the IC's generating resource. The metering equipment should be housed in a control cabinet or similar enclosure and must be accessible to PPL EU metering personnel.

6 Network Impacts

The Queue Project AE2-288 was evaluated as a 206.0 MW Capacity injection at **Sunbury 500 kV substation** in the PPL area. Project AE2-288 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-288 was studied with a commercial probability of 1.00. Potential network impacts were as follows:

Summer Peak Load Flow

7 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

8 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

9 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/DC	MW IMPACT
8519519	938390	AE1-058 TAP	230.0	PPL	208072	SIEG	230.0	PPL	1	PL:1A:P42:000922	breaker	628.0	101.9	104.16	AC	16.91
8519520	938390	AE1-058 TAP	230.0	PPL	208072	SIEG	230.0	PPL	1	PL:1A:P42:000923	breaker	628.0	101.9	104.16	AC	16.91
8519521	938390	AE1-058 TAP	230.0	PPL	208072	SIEG	230.0	PPL	1	PL:18:P42:000129	breaker	628.0	102.34	104.75	AC	17.87
8519522	938390	AE1-058 TAP	230.0	PPL	208072	SIEG	230.0	PPL	1	PL:08:P42:000130	breaker	628.0	102.34	104.75	AC	17.87
8519523	938390	AE1-058 TAP	230.0	PPL	208072	SIEG	230.0	PPL	1	PL:10:P42:100576	breaker	628.0	102.34	104.75	AC	17.87

10 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/DC	MW IMPACT
7366044	200021	SUNBURY	500.0	PJM	200009	JUNIATA	500.0	PJM	1	PL:28:P12:000080	operation	3112.0	109.98	114.11	AC	129.74
7366049	200021	SUNBURY	500.0	PJM	200009	JUNIATA	500.0	PJM	1	Base Case	operation	2707.0	96.15	100.32	AC	116.07
7366236	200022	SUSQHANA	500.0	PJM	200023	WESCOVLE	500.0	PJM	1	PL:08:P12:000083	operation	3112.0	104.74	107.71	AC	84.3

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
7366230	200023	WESCOVLE	500.0	PJM	200075	BREI	500.0	PJM	1	PL:08:P12:000083	operation	3112.0	105.15	108.21	AC	87.75
8519925	938390	AE1-058 TAP	230.0	PPL	208072	SIEG	230.0	PPL	1	PL:08:P12:000083	operation	628.0	102.29	104.7	AC	17.87

11 Stability and Reactive Power Requirement for Low Voltage Ride Through

Stability Analysis will be performed during the Facilities Study. **Additional reinforcements may be identified to mitigate stability concerns.**

12 Light Load Analysis

There are no mitigations required.

13 Affected Systems

None

14 System Reinforcements

ID	Index	Facility	Upgrade Description	Cost	AE2-288 Cost Allocation	Network Upgrade Number
8519519, 8519521, 8519520, 8519523, 8519522	4	AE1-058 TAP 230.0 kV - SIEG 230.0 kV Ckt 1	Reinforcement: Rebuild the AE1-058-Siegfried line segment of the Frackville – Siegfried 230kV Line Estimated Schedule: 72 Months Rating after the upgrade: 665/821/821	\$69,000,000	\$42,971,701	N6406
			TOTAL COST	\$69,000,000	\$42,971,701	

15 Flow Gate Details

The following indices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

15.1 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
8519523	938390	AE1-058 TAP	PPL	208072	SIEG	PPL	1	PL:10:P42:100576	breaker	628.0	102.34	104.75	AC	17.87

Bus #	Bus	MW Impact
208941	FISH CT	0.8778
208981	FOWH IPP	1.2072
208982	GLBT IPP	2.3214
209013	SCEN IPP	2.6961
209022	WHFR IPP (Deactivation : 03/01/20)	1.3480
211064	PSPA	0.6342
918431	AA1-057	1.4367
919512	AA2-008 E	4.6682
920711	AA2-182 C	72.0415
920712	AA2-182 E	3.9081
924291	AB2-074 C	3.6869
924292	AB2-074 E	4.6455
926081	AC1-087 C	0.2410
926082	AC1-087 E	0.3932
935071	AD1-143 C1	0.3938
935072	AD1-143 E1	2.3602
935081	AD1-143 C2	0.0162
935082	AD1-143 E2	0.3888
935091	AD1-143 C3	0.3592
935092	AD1-143 E3	2.1527
935101	AD1-143 C4	0.0148
935102	AD1-143 E4	0.3546
938391	AE1-058 C	86.8100
938392	AE1-058 E	86.8100
939712	AE1-202 E (Withdrawn : 11/04/2019)	0.2372
939891	AE1-225 C O1	0.8569
939892	AE1-225 E O1	0.9481
940561	AE2-042 C	4.1196
940562	AE2-042 E	2.0422
940721	AE2-059 C	0.7394
940722	AE2-059 E	1.0211
940941	AE2-084 C	0.7394
940942	AE2-084 E	1.0211
941161	AE2-110 C	0.7849
941162	AE2-110 E	1.0839
941171	AE2-111 C	0.7550
941172	AE2-111 E	1.0426
941371	AE2-133 C	0.7698
941372	AE2-133 E	1.0630
942281	AE2-241 C	0.7849
942282	AE2-241 E	1.0839
942561	AE2-271 C	4.8546

Bus #	Bus	MW Impact
942562	AE2-271 E	3.2364
942581	AE2-274	0.0507
942721	AE2-288	15.1899
942771	AE2-295 C	4.6486
942772	AE2-295 E	26.9692
DUCKCREEK	DUCKCREEK	0.2361
NEWTON	NEWTON	0.2848
CHILHOWEE	CHILHOWEE	0.0391
G-007	G-007	0.6300
TVA	TVA	0.3763
PRAIRIE	PRAIRIE	0.5373
O-066	O-066	5.2023
COFFEEN	COFFEEN	0.1089
EDWARDS	EDWARDS	0.1071
CHEOAH	CHEOAH	0.1096
TILTON	TILTON	0.1292
SANTEETLA	SANTEETLA	0.0323
SMITHLAND	SMITHLAND	0.0436
CBM-N	CBM-N	0.1178
COTTONWOOD	COTTONWOOD	0.4494
HAMLET	HAMLET	0.1345
BLUEG	BLUEG	1.0321
UNIONPOWER	UNIONPOWER	0.1694
CANNELTON	CANNELTON	0.0633
GIBSON	GIBSON	0.0426
CALDERWOOD	CALDERWOOD	0.1196
FARMERCITY	FARMERCITY	0.0728
TRIMBLE	TRIMBLE	0.1146
CATAWBA	CATAWBA	0.0801
NYISO	NYISO	0.4946

16 Contingency Definition

Contingency Name	Contingency Definition
PL:10:P42:100576	CONTINGENCY 'PL:10:P42:100576' /* AT JUNIATA 500SUB SUNBURY 500KV S.CB FAILED" DISCONNECT BRANCH FROM BUS 200009 TO BUS 200183 CKT 1 /* /* JUNIATA CAP 500 KV DISCONNECT BRANCH FROM BUS 200009 TO BUS 208004 CKT 1 /* /* JUNIATA 500-230 KV BK1 DISCONNECT BRANCH FROM BUS 200009 TO BUS 200021 CKT 1 /* /* JUNIATA-SUNBURY 500 KV (SHOULD BE FAULTED) END
PL:18:P42:000129	CONTINGENCY 'PL:18:P42:000129' /* JUNI-SUNB 500KV STUCK BREAKER CONNECTED TO TR2 DISCONNECT BRANCH FROM BUS 200009 TO BUS 208005 CKT 2 /* JUNIATA-JUNI BU2 500-230 DISCONNECT BRANCH FROM BUS 200009 TO BUS 200021 CKT 1 /* JUNIATA-SUNBURY 500 END
Base Case	
PL:08:P12:000083	CONTINGENCY 'PL:08:P12:000083' /* JUNI-SUNB 500KV LINE DISCONNECT BRANCH FROM BUS 200009 TO BUS 200021 CKT 1 /* JUNIATA-SUNBURY 500 END
PL:28:P12:000080	CONTINGENCY 'PL:28:P12:000080' /* SUSQ-WESC 500KV LINE DISCONNECT BRANCH FROM BUS 200022 TO BUS 200023 CKT 1 /* SUSQHANA-WESCOVLE 500 END
PL:08:P42:000130	CONTINGENCY 'PL:08:P42:000130' /* JUNI-SUNB 500KV STUCK BREAKER CONNECTED TO TR1 DISCONNECT BRANCH FROM BUS 200009 TO BUS 208004 CKT 1 /* JUNIATA-JUNI BU1 500-230 DISCONNECT BRANCH FROM BUS 200009 TO BUS 200021 CKT 1 /* JUNIATA-SUNBURY 500 END
PL:20:P12:000092	CONTINGENCY 'PL:20:P12:000092' /* BREI-WESC 500KV LINE DISCONNECT BRANCH FROM BUS 200023 TO BUS 200075 CKT 1 /* WESCOVLE-BREI 500 END
PL:1A:P42:000922	CONTINGENCY 'PL:1A:P42:000922' /* SUNBURY 500KV YARD 3N BF DISCONNECT BRANCH FROM BUS 200021 TO BUS 200009 CKT 1 /* /* JUNIATA-SUNBURY 500KV LINE DISCONNECT BRANCH FROM BUS 200021 TO BUS 208109 CKT 24 /* /* T24 END
PL:1A:P42:000923	CONTINGENCY 'PL:1A:P42:000923' /* SUNBURY 500KV YARD 3T BF DISCONNECT BRANCH FROM BUS 200021 TO BUS 200009 CKT 1 /* /* JUNIATA-SUNBURY 500KV LINE DISCONNECT BRANCH FROM BUS 200021 TO BUS 208109 CKT 25 /* /* T25 END

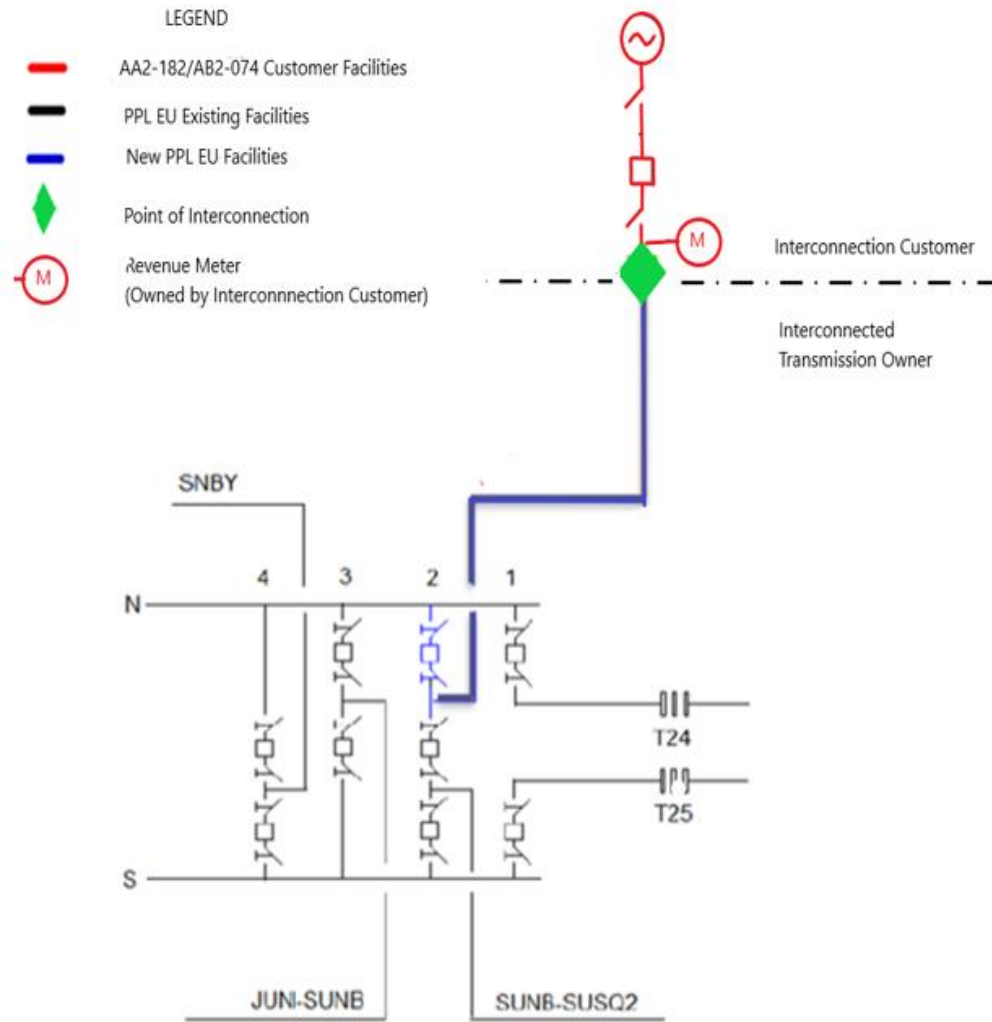
Short Circuit

17 Short Circuit

The following Breakers are over duty:

None

Attachment 1 Single Line



There is no additional Transmission Owner Interconnection Facilities associated with AE2-288 increase. Transmission Owner Scope will be completed under queue project AA2-182/AB2-074